

Interpretable Data Science Project

Customer Churn Prediction

Duc Manh NGUYEN

Minh Hoan TRAN

Qiushuang LIU



Business Problem & Our Approach

High churn costs

Acquiring new customers costs 5–10× more than retaining existing ones

Limited budget

Retention offers (discounts, free minutes) only reachable to top 25% at-risk customers

Interpretability required

Coefficients used operationally by agents – a black box breaks the workflow

Our Approach

1

EDA + Multicollinearity

Correlation matrix & VIF analysis; treat severe collinear features

2

Model Comparison

LR baseline → Tuned LR → Decision Tree → RF → XGBoost

3

Interpretability Suite

Coefficients, PDP, ICE, ALE, Global Surrogate, LIME, SHAP

4

Recommendation

Performance–interpretability trade-off with test set validation

Primary metric: ROC-AUC (ranking quality) • Business metric: Precision@Top25% (campaign efficiency) • Dataset: 10,000 customers, 18 features

Data Insights & Multicollinearity Treatment

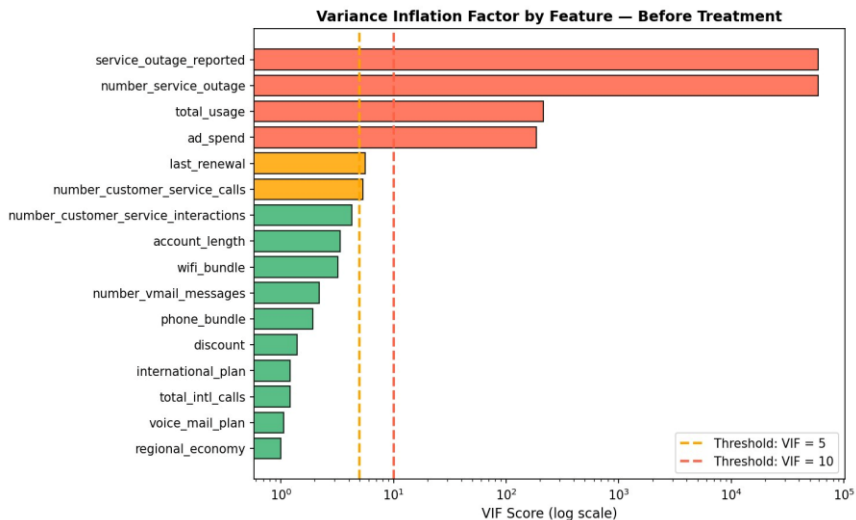
36.1%

Churn rate
(class imbalance noted)

Wifi Bundle subscribers: 18.9% churn
vs 41.1% non-subscribers – strongest
binary signal

Severe VIF detected

service_outage_reported /
number_service_outage: VIF > 50,000
total_usage / ad_spend: VIF > 100
calls / interactions: VIF ~ 5



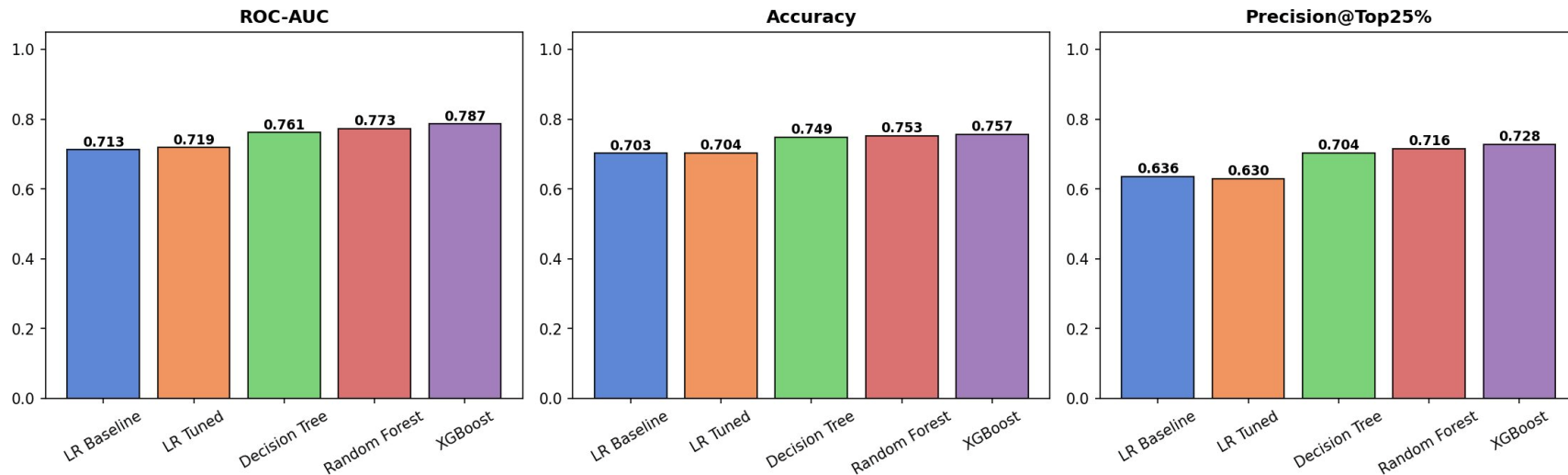
Treatment: 3 features dropped

Dropped	Reason
service_outage_reported	Near-perfect duplicate (VIF >50k)
ad_spend	Derived from total_usage (VIF >100)
interactions	Superset of calls (VIF ~5)

*Residual: total_usage & number_service_outage retain VIF ≈ 15
– retained as top predictors; disclosed as a deliberate business
trade-off.*

Model Comparison Results

Model Comparison — Validation Set



Interpretability tax

LR Tuned vs XGBoost: 9.8 pp gap in Precision@Top25% on validation

Real trade-off

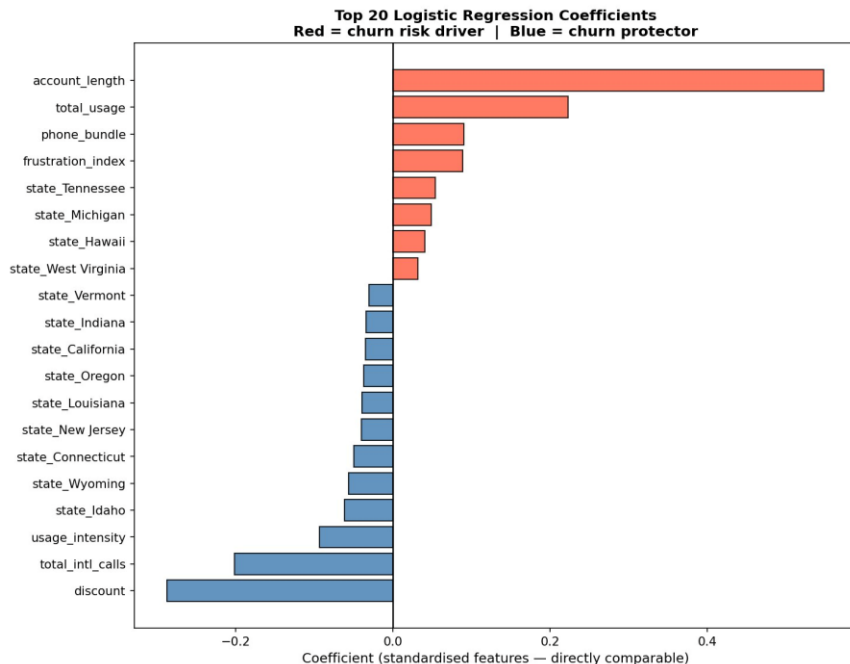
Decision Tree vs XGBoost: only 2.4 pp gap – shallow DT is dominant choice

Test set confirmed

LR: AUC 0.695 / P@25% 60.4% |
XGBoost: AUC 0.766 / P@25% 70.8%

Interpretability – Intrinsic Methods

Logistic Regression Coefficients



Key Findings

account_length (positive)

Longest-tenured customers have the highest churn risk – contra-intuitive; warrants business investigation

total_usage + frustration_index

Heavy users who also call service repeatedly are at acute risk

discount (strong negative)

Discounts are highly effective at retention – validates targeted pricing strategy

total_intl_calls (negative)

International callers are stickier – likely value the network specifically

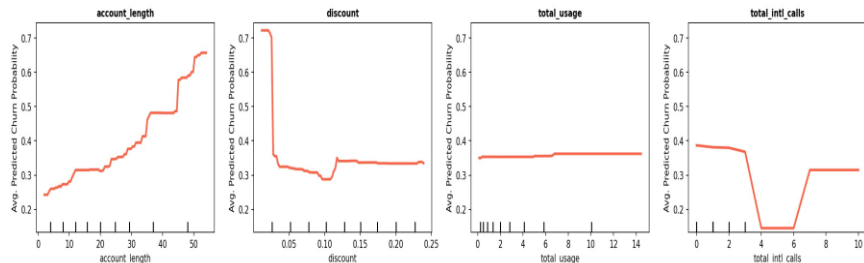
Features standardised – coefficients directly comparable across all variables

Interpretability - Global Post-hoc Methods

All post-hoc methods applied to XGBoost to illustrate what black-box deployment would require

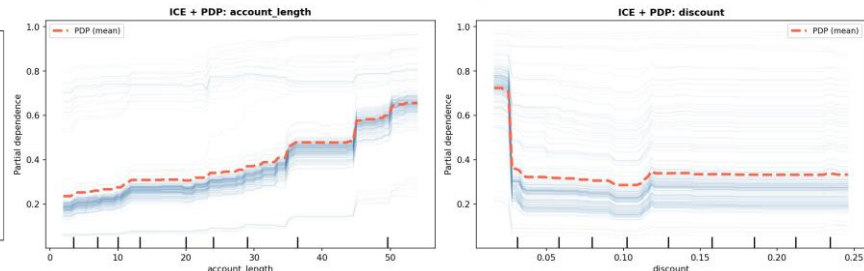
PDP - Average marginal effect

Partial Dependence Plots - XGBoost (top 4 features by importance)



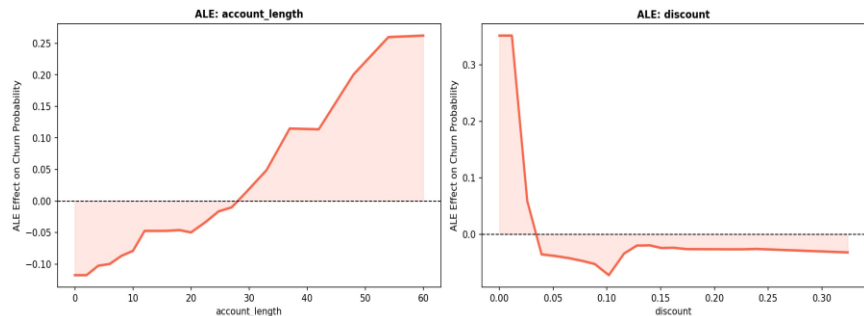
ICE - Per-customer curves reveal heterogeneity

ICE Plots - XGBoost (200 sampled customers)



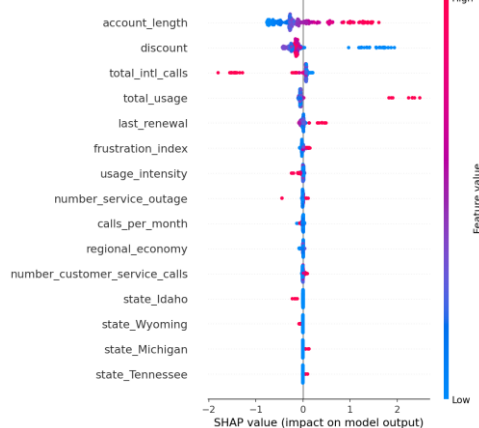
ALE - Unbiased under feature correlation

Accumulated Local Effects (ALE) - XGBoost



SHAP Beeswarm - Direction + magnitude

SHAP Beeswarm: Feature Impact and Direction



Interpretability - Local Explanations

Highest-Risk Customer - Tenure: 55mo | Usage: 24.75 days/mo | Intl Calls: 0 | Outages: 61 | Discount: 17.7% | SHAP score: 3.759

SHAP WATERFALL - Why this customer?



total_usage (+2.32) + account_length (+1.61) drive 93% of churn signal

LIME - Top Feature Attributions

Feature Condition	Attribution	Direction
account_length > 33	+0.26	↑ Churn
total_usage > 4.78	+0.08	↑ Churn
total_intl_calls ≤ 1	+0.07	↑ Churn
state_South Carolina ≤ 0	+0.05	↑ Churn
0.13 < discount ≤ 0.19	-0.03	↓ Protect
state_Rhode Island ≤ 0	-0.04	↓ Protect

ACTIONABLE INSIGHTS

Do NOT offer more discount

Discount is already applied (17.7%) and contributes only -0.03 against +4.3 of churn signal. Price alone won't retain this customer.

Address service quality directly

61 outages + frustration index 122 are active drivers. A service quality follow-up call is more effective than a price reduction.

Recognise loyalty

55 months is long-tenured. A loyalty reward or upgrade offer acknowledges tenure without devaluing pricing.

Recommendation & Business Actions

Model	Interpretability	AUC (Test)	P@25% (Test)	Verdict
LR Baseline	Full - coefficients	-	-	Replaced
Tuned LR ✓	Full - coefficients	0.695	60.4%	RECOMMENDED
Decision Tree	Full - IF-THEN rules	~0.76	~70.4%	Alt. if depth ≤ 5
Random Forest	Partial - importance	-	71.6%	Benchmark only
XGBoost	None - post-hoc	0.766	70.8%	Performance ceiling

Actionable Recommendations

Weekly churn scoring

Run model weekly; trigger retention for customers entering top decile for the first time

Outage = immediate risk signal

Auto-flag any customer with a service outage in past 30 days for proactive outreach

Targeted discounts

Discount coefficient confirms price sensitivity - deploy to model-flagged customers only, not broadly

Quarterly retraining

Trigger full retraining if AUC drops >3 points below benchmark

Conclusion



Multicollinearity identified and treated – 3 features dropped, VIF reduced from >50,000 to <16



Five models compared – XGBoost sets the performance ceiling; Decision Tree is the practical champion



Full interpretability suite delivered: coefficients, PDP, ICE, ALE, Global Surrogate, LIME, SHAP



Tuned Logistic Regression recommended: operationally interpretable, test AUC 0.695, Precision@Top25% 60.4%



Interpretability tax is real but quantified: XGBoost adds 10.4 pp precision at the cost of full black-box opacity

THANK YOU!

